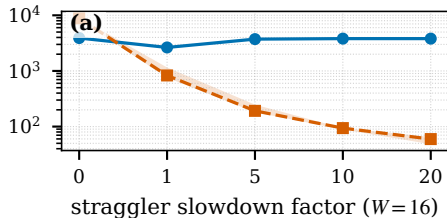


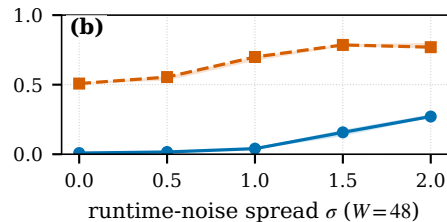
throughput (sims/s)

Asynchronous (ours)

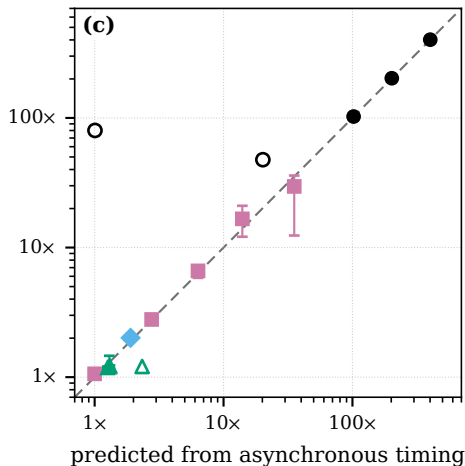
Synchronous baseline



worker idle fraction



measured against the twin



○ outside the model's domain

■ runtime heterogeneity ( $W=48$ )◆ Cellular Potts  $80^3$  ( $W=48$ )● persistent straggler ( $W=16$ )▲ Cellular Potts  $50^3$  ( $W=48-384$ )